

LOWMAC-PLUS

**Methylcobalamin 1000 mcg + Pyridoxine HCl 100 mg + Nicotinamide 100 mg + Thiamine HCl 100 mg + D-Panthenol 50 mg
Injection Kit**

1. Clinical Pharmacology

LOWMAC-PLUS is a neurotropic vitamin combination formulated to support neuronal metabolism, myelin repair, and peripheral nerve function. The formulation combines essential B-complex vitamins involved in neurotransmitter synthesis, axonal transport, cellular energy production, and regeneration of damaged neural tissue.

Methylcobalamin

Methylcobalamin is the neurologically active form of Vitamin B12 and plays a critical role in:

- Myelin sheath synthesis and regeneration
- Nucleic acid and protein synthesis
- Neuronal lipid metabolism
- Maintenance of axonal integrity
- Homocysteine metabolism

Methylcobalamin acts as a cofactor in methionine synthesis, promoting methylation reactions essential for neuronal repair. Deficiency may result in demyelination, impaired nerve conduction, and progressive neuropathic symptoms.

Experimental and clinical studies suggest that high-dose methylcobalamin may enhance nerve regeneration and improve nerve conduction velocity in peripheral neuropathies.

Pyridoxine Hydrochloride (Vitamin B6)

Pyridoxine is involved in:

- Neurotransmitter synthesis (GABA, serotonin, dopamine)
- Amino acid metabolism
- Sphingolipid synthesis required for nerve tissue

It supports normal neuronal function and may help reduce neuropathic symptoms associated with vitamin deficiency.

Thiamine Hydrochloride (Vitamin B1)

Thiamine is essential for:

- Carbohydrate metabolism
- ATP generation within neurons
- Axonal conduction
- Neural energy homeostasis

Thiamine deficiency is associated with peripheral neuropathy and impaired neuromuscular function.

Nicotinamide

Nicotinamide participates in cellular oxidation-reduction reactions through NAD/NADP-dependent pathways and contributes to:

- Cellular energy metabolism
- Tissue repair
- Neuroprotective cellular functions

D-Panthenol

D-Panthenol is a precursor of coenzyme A and contributes to:

- Cellular metabolic processes
 - Tissue regeneration
 - Maintenance of epithelial and neural tissue integrity
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2. Indications

LOWMAC-PLUS is indicated in conditions associated with deficiency or increased requirement of neurotropic vitamins, including:

- Peripheral neuropathy
- Diabetic neuropathy
- Alcoholic neuropathy

- Drug-induced neuropathy
 - Neuralgia and neuritis
 - Megaloblastic anemia due to Vitamin B12 deficiency
 - Nutritional deficiency states
 - Cervical and lumbar radiculopathy
 - Neuromuscular weakness associated with B-complex deficiency
 - Convalescence and chronic debilitating conditions
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3. Administration

Route of Administration

- Intramuscular (I.M.)
- Intravenous (I.V.) administration under medical supervision

Recommended Dosage

As directed by the physician depending on the severity of deficiency and clinical condition.

Intramuscular Administration

- Deep intramuscular injection is generally preferred for routine therapy.

Intravenous Administration

- Slow intravenous administration may be used in selected clinical settings under supervision.
- Rapid intravenous injection should be avoided.

Handling Precautions

- Inspect visually before administration.
 - Do not use if particulate matter or discoloration is observed.
 - Maintain aseptic precautions during preparation and administration.
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4. Scientific Research Insight

Recent clinical and experimental studies have highlighted the therapeutic importance of high-dose methylcobalamin in chronic neuropathic disorders.

High-dose Vitamin B12 therapy has demonstrated potential benefits in:

- Enhancement of nerve regeneration
- Improvement in nerve conduction velocity
- Reduction in paresthesia and burning pain
- Support of remyelination processes

In diabetic neuropathy and chronic peripheral nerve injury, methylcobalamin has been associated with symptomatic improvement in sensory disturbances and functional recovery. Mechanistic studies suggest that methylcobalamin may facilitate protein synthesis within injured neurons and promote axonal transport mechanisms necessary for nerve repair.

Clinical observations further indicate that combination therapy with Vitamins B1, B6, and B12 may provide synergistic support to neuronal metabolism and reduce oxidative stress associated with chronic neuropathic conditions.

5. Contraindications

LOWMAC-PLUS is contraindicated in patients with:

- Known hypersensitivity to methylcobalamin, cobalt, or any component of the formulation
 - Hypersensitivity to thiamine, pyridoxine, nicotinamide, or D-panthenol
 - Polycythemia vera (relative contraindication in Vitamin B12 therapy)
 - Leber's hereditary optic neuropathy
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6. Warnings and Precautions

Use with caution in:

- Patients with renal impairment
- Patients receiving long-term high-dose pyridoxine therapy
- Individuals with hypersensitivity disorders
- Patients with severe hepatic dysfunction

Prolonged excessive intake of pyridoxine may rarely be associated with sensory neuropathy.

Parenteral administration should be performed under medical supervision due to the possibility of hypersensitivity reactions.

7. Adverse Reactions

The following adverse reactions have been reported with neurotropic vitamin preparations:

General Disorders

- Injection site pain
- Local irritation
- Induration

Gastrointestinal Disorders

- Nausea
- Vomiting
- Mild gastrointestinal discomfort

Nervous System Disorders

- Headache
- Dizziness

Dermatological Reactions

- Rash
- Pruritus
- Urticaria

Hypersensitivity Reactions

- Allergic reactions
- Rare anaphylactoid reactions following parenteral administration

Metabolic Effects

- Flushing (occasionally with nicotinamide-containing preparations)

Patients should be monitored for signs of hypersensitivity during and after administration.

8. Storage

Store below 25°C in a dry place, protected from light.

Do not freeze.

Keep out of reach of children.

9. Dosage Form

Injection Kit for parenteral administration.